# Distributed Model Predictive Control for Energy Management of a PV–Battery–Hydrogen DC Microgrid

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***Abstract*—This paper presents a distributed model predictive control (DMPC) framework for energy management in a DC microgrid integrating solar photovoltaic (PV) generation, battery energy storage, and hydrogen-based energy systems. The proposed system incorporates a 36 kW PV array, a battery storage unit, a 10 kW proton-exchange membrane fuel cell (PEMFC), and a 10 kW alkaline electrolyzer (AE) interconnected through a common 300 V DC bus. A distributed control strategy is developed to ensure coordinated operation of subsystems under varying load and generation conditions. The control objective includes DC bus voltage regulation, power balance, and optimal energy utilization. Simulation results demonstrate improved stability, efficient energy distribution, and effective handling of renewable intermittency.**

***Index Terms*—DC microgrid, Distributed MPC, Solar PV, Hydrogen energy, Fuel cell, Electrolyzer, Energy management, Grey Wolf Optimizer.**

## I. Introduction

The increasing penetration of solar photovoltaic (PV) systems in modern power systems necessitates advanced energy management strategies to handle intermittency and ensure reliable operation. Hybrid energy storage systems combining batteries and hydrogen technologies provide a promising solution for both short-term and long-term energy balancing.

Conventional rule-based energy management approaches suffer from limited adaptability under dynamic conditions. Model predictive control (MPC), particularly in distributed form, offers improved flexibility and scalability by allowing each subsystem to optimize its own local objective while exchanging only a minimal set of communication variables with neighbouring controllers.

This paper proposes a distributed economic MPC (DEMPC)-based energy management system for a DC microgrid integrating PV, battery, PEMFC, and alkaline electrolyzer. The main contributions are:

- Development of a detailed nonlinear dynamic model of the hybrid PV–battery–hydrogen DC microgrid at the converter level.
- Formulation of a distributed economic MPC strategy with subsystem-specific droop-based cost functions and a supervisory energy management system (EMS).
- Validation under dynamic load and irradiance scenarios demonstrating DC bus voltage regulation and coordinated hydrogen system operation.

## II. System Description

The DC microgrid consists of a PV array, a battery energy storage system (BESS), a PEMFC stack, an alkaline electrolyzer, and a variable DC load connected through a common DC bus of nominal voltage $V_{dc}^* = 300\,\text{V}$. Each source and storage element is interfaced via a dedicated DC–DC converter. The overall system power balance is:

$$P_{pv} + P_{fc} + P_{bat} = P_{load} + P_{ae} + P_{loss} \tag{1}$$

where $P_{pv}$, $P_{fc}$, $P_{bat}$, $P_{ae}$, and $P_{load}$ denote the PV, fuel cell, battery, electrolyzer, and load powers, respectively, and $P_{loss}$ accounts for converter losses (neglected in the ideal model).

## III. Component Modeling

This section presents the dynamic modeling of all major components. The system is modeled at the converter level to capture switching dynamics and nonlinear behavior, which is essential for the predictive control implementation.

### A. Photovoltaic Array

The PV array consists of $N_s = 5$ modules in series and $N_{sh} = 20$ parallel strings. Each module is modeled using the single-diode equivalent circuit with both series resistance $R_s$ and shunt resistance $R_{sh}$. For an array of $N_s \times N_{sh}$ modules the implicit current equation is:

$$\begin{aligned} I_{pv} = N_{sh}\, I_{ph} - N_{sh}\, I_0 \left( \exp\left( \frac{V_{pv} + I_{pv} R_{s,eq}}{a N_s V_t} \right) - 1 \right) \\ - \frac{V_{pv} + I_{pv} R_{s,eq}}{R_{sh,eq}} \end{aligned} \tag{2}$$

where the array-equivalent series and shunt resistances are:

$$R_{s,eq} = \frac{N_s}{N_{sh}} R_s, \qquad R_{sh,eq} = \frac{N_s}{N_{sh}} R_{sh} \tag{3}$$

and $V_t = kT/q$ is the cell thermal voltage at operating temperature $T$, $a$ is the diode ideality factor, and $N_{cell}$ is the number of cells per module. Because (**??**) is implicit in $I_{pv}$, it is solved at each time step using Newton–Raphson iteration.

The photocurrent depends on irradiance $G$ and temperature $T$:

$$I_{ph} = \frac{G}{1000} (I_{scn} + k_i\,(T - T_n)) \tag{4}$$

where $I_{scn}$ is the short-circuit current at standard test conditions (STC: $G_n = 1000\,\mathrm{W/m^2}$, $T_n = 25\,^\circ\mathrm{C}$) and $k_i$ is the temperature coefficient.

The saturation current at operating temperature is:

$$I_0 = I_{on}\left(\frac{T}{T_n}\right)^3 \exp\left[\frac{qE_{gap}}{ka}\left(\frac{1}{T_n} - \frac{1}{T}\right)\right] \tag{5}$$

where the nominal saturation current at STC is:

$$I_{on} = \frac{I_{scn}}{\exp\left(\dfrac{V_{ocn}}{aV_{t,n}}\right) - 1} \tag{6}$$

The PV array output power is:

$$P_{pv} = V_{pv}\, I_{pv} \tag{7}$$

### B. PV Boost Converter

The PV array is interfaced with the DC bus through a unidirectional boost converter. The converter introduces two dynamic states: the PV-side capacitor voltage $v_{pv}$ and the boost inductor current $i_L$. Their state equations are:

$$C_s\,\frac{dv_{pv}}{dt} = i_{pv}(v_{pv}) - i_L \tag{8}$$

$$L_s\,\frac{di_L}{dt} = v_{pv} - (1 - d_s)\,V_{dc} \tag{9}$$

The current injected into the DC bus is:

$$i_s = (1 - d_s)\,i_L \tag{10}$$

where $d_s \in [0, 1]$ is the boost converter duty cycle and $C_s$, $L_s$ are the filter capacitance and inductance, respectively.

### C. Alkaline Electrolyzer Stack

The alkaline electrolyzer (AE) stack consists of $N_{ae} = 30$ cells in series. The per-cell terminal voltage combines the reversible voltage, an ohmic drop, and a Tafel overvoltage:

$$\begin{aligned} V_{ae}(i_{ae}) = N_{ae}\Bigg[\, & U_{rev} + \frac{r_1 + r_2\,T_{ae}}{A}\,i_{ae} \\ & + t_e \ln\left(\frac{t_1 + t_2/T_{ae} + t_3/T_{ae}^2}{A}\,i_{ae} + 1\right)\Bigg] \end{aligned} \tag{11}$$

where $U_{rev} = 1.229\,\mathrm{V}$ is the reversible cell voltage, $r_1$, $r_2$ are ohmic resistance coefficients, $t_e$, $t_1$, $t_2$, $t_3$ are Tafel overvoltage parameters, $A$ is the active cell area, and $T_{ae}$ is the cell temperature.

The AE stack power consumed is:

$$P_{ae} = V_{ae}(i_{ae})\,i_{ae} \tag{12}$$

The Faraday (current) efficiency, which accounts for parasitic current losses, is modeled as:

$$\eta_F = \frac{(i_{ae}/A)^2}{f_1 + (i_{ae}/A)^2}\,f_2 \tag{13}$$

where $f_1$ and $f_2$ are empirical efficiency parameters. The hydrogen production rate is then:

$$\dot{N}_{H_2} = \frac{\eta_F\,N_{ae}\,i_{ae}}{2F} \tag{14}$$

where $F = 96485.3\,\mathrm{C/mol}$ is the Faraday constant.

### D. AE Buck Converter

The electrolyzer is supplied from the DC bus through a unidirectional buck converter. The inductor current dynamics are:

$$L_{ae}\,\frac{di_{ae}}{dt} = d_{ae}\,V_{dc} - V_{ae}(i_{ae}) \tag{15}$$

The current drawn from the DC bus is:

$$i_a = d_{ae}\,i_{ae} \tag{16}$$

where $d_{ae} \in [0, 1]$ is the buck converter duty cycle and $L_{ae}$ is the converter inductance.

### E. PEMFC Stack

The proton-exchange membrane fuel cell (PEMFC) stack consists of $N_{pe} = 300$ cells in series operating at temperature $T_{pe} = 323.15\,\mathrm{K}$. The stack output voltage is composed of the thermodynamic Nernst voltage less three polarization losses:

$$V_{pe}(i_{pe}) = N_{pe}(E_{nernst} + v_{act} + v_{ohm} + v_{con}) \tag{17}$$

*Nernst EMF:* The open-circuit thermodynamic voltage per cell is:

$$\begin{aligned} E_{nernst} = {} & 1.229 - 8.45\times10^{-4}(T_{pe} - 298.15) \\ & + 4.3085\times10^{-5}\,T_{pe}(\ln P_{H_2} + 0.5\ln P_{O_2}) \end{aligned} \tag{18}$$

where $P_{H_2}$ and $P_{O_2}$ are the hydrogen and oxygen partial pressures in atm.

*Activation polarization:*

$$\begin{aligned} v_{act} = {} & -0.9514 + 3.12\times10^{-3}T_{pe} + 7.4\times10^{-5}T_{pe}\ln C_{O_2} \\ & - 1.87\times10^{-4}T_{pe}\ln i_{pe} \end{aligned} \tag{19}$$

where $C_{O_2}$ is the oxygen concentration at the cathode–catalyst interface:

$$C_{O_2} = \frac{P_{O_2}}{5.08\times10^6\,\exp(-498/T_{pe})} \tag{20}$$

*Ohmic polarization:* The Nafion membrane resistivity is:

$$r_m = \frac{181.6\left[1 + 0.03J + 0.062(T_{pe}/303)^2 J^{2.5}\right]}{(\lambda - 0.634 - 3J)\exp[4.18(T_{pe} - 303)/T_{pe}]} \tag{21}$$

where $J = i_{pe}/A_{pe}$ is the current density ($\mathrm{A/cm^2}$) and $\lambda$ is the membrane water content. The ohmic loss per cell is:

$$v_{ohm} = -i_{pe}\left(\frac{r_m\,l_m}{A_{pe}} + R_c\right) \tag{22}$$

where $l_m$ is the membrane thickness and $R_c$ is the contact resistance.

*Concentration polarization:*

$$v_{con} = 0.016\ln\left(1 - \frac{i_{pe}}{A_{pe}\,J_{max}}\right) \tag{23}$$

where $J_{max}$ is the maximum current density.

*Stack power and hydrogen consumption:*

$$P_{pe} = V_{pe}(i_{pe})\, i_{pe} \tag{24}$$

$$\dot{q}_{H_2} = \frac{N_{pe}\, i_{pe}}{2F} \tag{25}$$

### F. PEMFC Boost Converter

The fuel cell is connected to the DC bus through a unidirectional boost converter. The inductor current dynamics are:

$$L_{pe}\,\frac{di_{pe}}{dt} = V_{pe}(i_{pe}) - (1 - d_{pe})\, V_{dc} \tag{26}$$

The current injected into the DC bus is:

$$i_p = (1 - d_{pe})\, i_{pe} \tag{27}$$

where $d_{pe} \in [0, 1]$ is the boost converter duty cycle.

### G. Battery Energy Storage System

The battery is modeled using a linear internal resistance model. The terminal voltage is:

$$V_{bat} = V_{nom} - R_{int}\, I_{bat} \tag{28}$$

where $V_{nom} = 240\,\mathrm{V}$ is the nominal voltage and $R_{int}$ is the internal resistance. The battery output power is:

$$P_{bat} = V_{bat}\, I_{bat} \tag{29}$$

The state-of-charge dynamics follow from Coulomb counting:

$$\frac{d\,\mathrm{SOC}}{dt} = -\frac{I_{bat}}{C_{bat}} \tag{30}$$

where $C_{bat} = C_{Ah} \times 3600\,\mathrm{As}$ is the battery capacity ($C_{Ah} = 100\,\mathrm{Ah}$). The SOC is constrained to safe limits:

$$\mathrm{SOC}_{min} \le \mathrm{SOC} \le \mathrm{SOC}_{max} \tag{31}$$

### H. Battery Bidirectional Converter

The battery is interfaced through a bidirectional DC–DC buck-boost converter. The inductor current dynamics are:

$$L_b\,\frac{di_b}{dt} = V_{bat} - (1 - d_b)\, V_{dc} \tag{32}$$

The current injected into (or drawn from) the DC bus is:

$$i_{b,dc} = (1 - d_b)\, i_b \tag{33}$$

where $d_b \in [0, 1]$ is the converter duty cycle. Positive $i_b$ denotes discharging; negative $i_b$ denotes charging.

### I. DC Bus Dynamics

The DC bus voltage dynamics are governed by the capacitor KCL:

$$C_{dc}\,\frac{dV_{dc}}{dt} = i_s + i_p + i_{b,dc} - i_a - i_l \tag{34}$$

The load is modeled as a constant power load (CPL), giving a nonlinear current term:

$$i_l = \frac{P_{load}}{V_{dc}} \tag{35}$$

### J. Complete State-Space Representation

Combining all subsystem models, the full system state vector is:

$$x = \begin{bmatrix} v_{pv} & i_L & i_{ae} & i_{pe} & V_{dc} & i_b & \mathrm{SOC} \end{bmatrix}^T \in \mathbb{R}^7 \tag{36}$$

The control input vector (converter duty cycles) is:

$$u = \begin{bmatrix} d_s & d_{ae} & d_{pe} & d_b \end{bmatrix}^T \in [0, 1]^4 \tag{37}$$

and the disturbance vector is:

$$w = \begin{bmatrix} G & T & P_{load} \end{bmatrix}^T \tag{38}$$

The complete nonlinear state equations are:

$$\begin{aligned}
C_s\,\dot{v}_{pv} &= i_{pv}(v_{pv}, G, T) - i_L \\
L_s\,\dot{i}_L &= v_{pv} - (1 - d_s)\, V_{dc} \\
L_{ae}\,\dot{i}_{ae} &= d_{ae}\, V_{dc} - V_{ae}(i_{ae}) \\
L_{pe}\,\dot{i}_{pe} &= V_{pe}(i_{pe}) - (1 - d_{pe})\, V_{dc} \\
C_{dc}\,\dot{V}_{dc} &= i_s + i_p + i_{b,dc} - i_a - P_{load}/V_{dc} \\
L_b\,\dot{i}_b &= V_{bat} - (1 - d_b)\, V_{dc} \\
\dot{\mathrm{SOC}} &= -I_{bat}/C_{bat}
\end{aligned} \tag{39}$$

This nonlinear system forms the prediction model used in the DEMPC formulation described in Section **??**.

## IV. Energy Management System

A supervisory energy management system (EMS) operates at the highest level of the control hierarchy and determines which hydrogen subsystems are permitted to operate at each sampling instant. The EMS acts as a discrete-event coordinator: it does not compute duty cycles directly, but rather issues binary activation flags that enable or disable the alkaline electrolyzer (AE) and the PEMFC via virtual contactors. This architecture decouples the slow energy-scheduling decisions from the fast converter-level MPC loops.

### A. Net Power Balance

At every sampling instant $k$, the EMS evaluates the available power balance using the PV maximum power $P_{pv,max}(k)$, computed by sweeping the PV array model across its voltage range and extracting the global peak:

$$P_{net}(k) = P_{pv,max}(k) - P_{load}(k) \tag{40}$$

A positive value of $P_{net}$ indicates a generation surplus that must be absorbed; a negative value indicates a generation deficit that must be compensated. The battery handles imbalances within its power rating $P_{bat}^{max}$; beyond this limit, or when the battery SOC reaches a boundary, the hydrogen subsystems are engaged.

### B. Activation Logic

The EMS generates two binary activation flags, $\zeta_a \in \{0,1\}$ for the electrolyzer and $\zeta_p \in \{0,1\}$ for the fuel cell, according to the following priority-ordered logic:

**Step 1 — Default state.** Both flags are initialised to zero at every time step:

$$\zeta_a(k) = 0, \qquad \zeta_p(k) = 0 \tag{41}$$

**Step 2 — Electrolyzer activation.** The electrolyzer is enabled ($\zeta_a = 1$) if either of the following surplus conditions holds:

$$\zeta_a = 1 \quad \text{if} \quad \begin{array}{l} P_{net} > P_{bat}^{max} \\ \text{or} \\ \big(P_{net} > 0 \ \wedge \ \mathrm{SOC} \geq \mathrm{SOC}_{max}\big) \end{array} \tag{42}$$

The first condition handles cases in which the generation surplus exceeds the battery charging power limit and would otherwise cause the DC bus voltage to rise uncontrollably. The second condition prevents the battery from being overcharged when it is already near full capacity but a surplus still exists.

**Step 3 — Fuel cell activation.** The fuel cell is enabled ($\zeta_p = 1$) if either of the following deficit conditions holds:

$$\zeta_p = 1 \quad \text{if} \quad \begin{array}{l} P_{net} < -P_{bat}^{max} \\ \text{or} \\ \big(P_{net} < 0 \ \wedge \ \mathrm{SOC} \leq \mathrm{SOC}_{min}\big) \end{array} \tag{43}$$

The first condition handles cases in which the deficit exceeds the battery discharge power limit. The second condition prevents deep discharge when the battery SOC has reached its lower bound.

The numerical values of all EMS thresholds are summarised in Table **??**.

TABLE I
EMS THRESHOLD PARAMETERS

| Parameter | Symbol | Value |
|---|---|---|
| Battery power limit | $P_{bat}^{max}$ | 5 kW |
| Minimum SOC | $\mathrm{SOC}_{min}$ | 0.20 |
| Maximum SOC | $\mathrm{SOC}_{max}$ | 0.90 |

### C. Mutual Exclusivity and Priority

The two activation conditions (**??**) and (**??**) are evaluated independently, which means that in pathological operating conditions — for example, a large surplus coinciding with a depleted battery — both $\zeta_a$ and $\zeta_p$ could simultaneously equal one. In normal operation this is physically inconsistent: it would require the electrolyzer to absorb power while the fuel cell simultaneously produces power from the same hydrogen store.

To prevent simultaneous activation, an explicit priority rule is enforced after Steps 2 and 3:

$$\text{if } \zeta_a = 1 \ \Rightarrow \ \zeta_p = 0 \tag{44}$$

Electrolyzer absorption is given priority over fuel-cell dispatch. This choice reflects the physical preference to store surplus energy as hydrogen rather than to simultaneously consume and produce it. Under this rule, the feasible flag combinations and their physical interpretations are:

$$(\zeta_a,\, \zeta_p) \in \big\{(0,0),\ (1,0),\ (0,1)\big\} \tag{45}$$

corresponding to *idle*, *electrolyzer active*, and *fuel cell active* modes, respectively.

### D. Virtual Contactor Enforcement

When a subsystem is deactivated ($\zeta_a = 0$ or $\zeta_p = 0$), a virtual contactor is opened: the corresponding converter duty cycle is set to zero and the associated inductor current state is forced to zero at the end of every integration step:

$$\zeta_a = 0 \ \Rightarrow \ d_{ae} = 0, \quad i_{ae} \leftarrow 0 \tag{46}$$

$$\zeta_p = 0 \ \Rightarrow \ d_{pe} = 0, \quad i_{pe} \leftarrow 0 \tag{47}$$

This hard clamping avoids spurious current dynamics in the inductor states of inactive converters and ensures that the DEMPC prediction models of the active subsystems receive consistent communicated values (zero power and zero bus current) from inactive subsystems.

The complete EMS decision procedure executed at each sampling instant is summarised in Algorithm **??**.

**Algorithm 1:** Supervisory EMS Decision Procedure

1) Compute $P_{pv,max}(k)$ via MPP sweep of PV array model.
2) Evaluate $P_{net}(k) = P_{pv,max}(k) - P_{load}(k)$.
3) Set $\zeta_a \leftarrow 0$, $\zeta_p \leftarrow 0$.
4) **If** $P_{net} > P_{bat}^{max}$ **or** ($P_{net} > 0$ **and** $\mathrm{SOC} \geq \mathrm{SOC}_{max}$): set $\zeta_a \leftarrow 1$.
5) **If** $P_{net} < -P_{bat}^{max}$ **or** ($P_{net} < 0$ **and** $\mathrm{SOC} \leq \mathrm{SOC}_{min}$): set $\zeta_p \leftarrow 1$.
6) **If** $\zeta_a = 1$: set $\zeta_p \leftarrow 0$ (electrolyzer priority rule).
7) Pass $(\zeta_a, \zeta_p)$ to local DEMPC controllers and enforce virtual contactors (**??**)–(**??**).

## V. DISTRIBUTED ECONOMIC MODEL PREDICTIVE CONTROL

### A. Architecture Overview

The DEMPC framework decomposes the centralized problem into four independent local controllers: PV subsystem (PVS), AE subsystem (AES), PEMFC subsystem (PEMFCS), and BESS. At each sampling instant $T_s$, each local controller:

1) Receives communicated power and current signals from the other subsystems (from the previous time step).
2) Solves its own finite-horizon optimal control problem over a prediction horizon $N_p$.
3) Applies the first element of the optimal duty-cycle sequence.

4) Broadcasts its power and bus current to the other controllers.

The DC bus voltage reference is updated with a reference smoothing filter:

$$V_{dc,ref}(k+1) = V_{dc,ref}(k) + \frac{1}{N_s}(V_{dc}^* - V_{dc,ref}(k)) \quad (48)$$

where $N_s = 10$ is the smoothing horizon and $V_{dc}^* = 300\,\text{V}$.

### B. Local Cost Function — PV Subsystem (PVS)

The PVS controller minimizes a cost that tracks the maximum power point (MPPT), regulates the DC bus voltage, and tracks the MPPT inductor current:

$$J_{pvs} = \sum_{l=1}^{N_p}\Big[w_{s2}\,\ell_{s2}^l + w_{s3}\,\ell_{s3}^l + w_{s4}\,\ell_{s4}^l\Big] \quad (49)$$

where the normalized stage costs at prediction step $l$ are:

$$\begin{cases} \ell_{s2}^l = \left(\dfrac{P_{pv}^l - P_{max}}{\max(P_{max},1)}\right)^2 & (50)\\ \ell_{s3}^l = \left(\dfrac{V_{dc}^l - V_{dc,ref}}{V_{dc,ref}}\right)^2 & (51)\\ \ell_{s4}^l = \left(\dfrac{i_L^l - i_{L,ref}}{\max(i_{L,ref},1)}\right)^2 & (52)\end{cases}$$

with weights $w_{s2} = 800$, $w_{s3} = 20000$, $w_{s4} = 400$. The inductor current reference $i_{L,ref}$ is set to the MPP current $I_{mpp}$ computed from a 500-point voltage sweep.

### C. Local Cost Function — AE Subsystem (AES)

The AES controller drives the electrolyzer power toward a surplus absorption target while regulating the bus voltage:

$$J_{aes} = \sum_{l=1}^{N_p}\Big[w_{a1}\,\ell_{a1}^l + w_{a2}\,\ell_{a2}^l\Big] \quad (53)$$

$$\begin{cases} \ell_{a1}^l = \left(\dfrac{P_{ae}^l - P_{tgt,a}}{\max(P_{tgt,a},1)}\right)^2 & (54)\\ \ell_{a2}^l = \left(\dfrac{V_{dc}^l - V_{dc,ref}}{V_{dc,ref}}\right)^2 & (55)\end{cases}$$

with weights $w_{a1} = 400$, $w_{a2} = 20000$. The surplus absorption target incorporates a voltage-droop correction:

$$P_{tgt,a} = \max\Big(P_{pv} + P_{fc} - P_{load} + K_{vdc}(V_{dc} - V_{dc,ref}),\ 0\Big) \quad (56)$$

where $K_{vdc} = 400\,\text{W/V}$ is the droop gain.

### D. Local Cost Function — PEMFC Subsystem (PEMFCS)

The PEMFCS controller drives the fuel cell toward a deficit compensation target:

$$J_{pemfcs} = \sum_{l=1}^{N_p}\Big[w_{p1}\,\ell_{p1}^l + w_{p2}\,\ell_{p2}^l\Big] \quad (57)$$

$$\begin{cases} \ell_{p1}^l = \left(\dfrac{P_{pe}^l - P_{tgt,p}}{\max(P_{tgt,p},1)}\right)^2 & (58)\\ \ell_{p2}^l = \left(\dfrac{V_{dc}^l - V_{dc,ref}}{V_{dc,ref}}\right)^2 & (59)\end{cases}$$

with weights $w_{p1} = 400$, $w_{p2} = 20000$. The deficit compensation target is:

$$P_{tgt,p} = \max\Big(P_{load} + P_{ae} - P_{pv} - K_{vdc}(V_{dc} - V_{dc,ref}),\ 0\Big) \quad (60)$$

### E. Local Cost Function — BESS

The BESS controller minimizes a composite cost penalizing voltage deviation, power imbalance, duty-cycle variation, and constraint violations:

$$J_{bess} = \sum_{l=1}^{N_p}\Big[w_v\,\ell_v^l + w_{du}\,\ell_{du}^l + w_p\,\ell_p^l + w_{lim}\,\ell_{lim}^l + w_{soc}\,\ell_{soc}^l\Big] \quad (61)$$

$$\begin{cases} \ell_v^l = \left(\dfrac{V_{dc}^l - V_{dc,ref}}{V_{dc,ref}}\right)^2 & (62)\\ \ell_{du}^l = (d_b^l - d_b^{l-1})^2 & (63)\\ \ell_p^l = \left(\dfrac{P_{pv} + P_{pe} + P_{bat}^l - P_{ae} - P_{load}}{\max(P_{load},1)}\right)^2 & (64)\\ \ell_{lim}^l = \left(\dfrac{\max(0,\ |P_{bat}^l| - P_{bat}^{max})}{P_{bat}^{max}}\right)^2 & (65)\\ \ell_{soc}^l = \left(\dfrac{\Psi_{soc}(P_{bat}^l,\ \text{SOC}^l)}{P_{bat}^{max}}\right)^2 & (66)\end{cases}$$

with weights $w_v = 20000$, $w_{du} = 100$, $w_p = 5000$, $w_{lim} = w_{soc} = 50000$, and $P_{bat}^{max} = 5\,\text{kW}$. The SOC violation penalty is:

$$\Psi_{soc} = \begin{cases} |P_{bat}^l| & \text{if SOC}^l \geq 0.9 \text{ and } P_{bat}^l < 0\\ |P_{bat}^l| & \text{if SOC}^l \leq 0.2 \text{ and } P_{bat}^l > 0\\ 0 & \text{otherwise}\end{cases} \quad (67)$$

### F. Optimization and Prediction

At each sampling instant, each local controller optimizes over a sequence of $N_p$ duty cycles. The local prediction model integrates the subsystem state equations using a forward-Euler scheme with four sub-steps per sampling interval $T_s = 40\,\mu\text{s}$. The duty cycles of all other subsystems are frozen at their communicated values during the local prediction (distributed approximation).

Each optimization problem takes the form:

$$\min_{d^{[1:N_p]}} \; J_i(d^{[1:N_p]}) \quad \text{s.t.} \quad d^l \in [\underline{d}_i,\, \overline{d}_i],\ \forall l \tag{68}$$

For the baseline controller, the feasible duty-cycle space is discretized into a 7-point adaptive grid centered on the previous duty cycle, and all $7^{N_p}$ sequences are enumerated. For the Grey Wolf Optimizer (GWO) variant, the same cost function is minimized using a population-based metaheuristic, reducing computational complexity while retaining solution quality.

## VI. Simulation Results

Simulation results demonstrate:

- Stable DC bus voltage regulation to 300 V under step and ramp disturbances in both load and irradiance.
- Near-MPPT PV power extraction across all operating conditions.
- Smooth and coordinated activation of the electrolyzer during surplus and the fuel cell during deficit, governed by the EMS logic.
- Effective battery SOC management within the $[0.20,\, 0.90]$ safe operating window.

## VII. Conclusion

A distributed economic MPC framework for a PV–battery–hydrogen DC microgrid has been presented. Detailed nonlinear models of all power conversion stages, including the full single-diode PV array, alkaline electrolyzer electrochemistry, and PEMFC polarization losses, were incorporated directly into the local prediction models. The EMS layer coordinates hydrogen system activation based on power surplus/deficit and battery SOC. Simulation results confirm stable voltage regulation, MPPT tracking, and coordinated energy dispatch under dynamic operating conditions.

Future work includes experimental validation, real-time embedded implementation, and comparison with centralized MPC benchmarks.

## Appendix

## References

TABLE II
DC Microgrid Component Parameters

| Parameter | Symbol | Value |
|---|---|---|
| *PV Array* | | |
| Series modules | $N_s$ | 5 |
| Parallel strings | $N_{sh}$ | 20 |
| Cells per module | $N_{cell}$ | 72 |
| Open-circuit voltage | $V_{ocn}$ | 47.6 V |
| Short-circuit current | $I_{scn}$ | 9.99 A |
| Series resistance | $R_s$ | 0.20 Ω |
| Shunt resistance | $R_{sh}$ | 250.0 Ω |
| Ideality factor | $a$ | 1.30 |
| *PV Boost Converter* | | |
| Capacitance | $C_s$ | 2000 $\mu$F |
| Inductance | $L_s$ | 1 mH |
| *Alkaline Electrolyzer* | | |
| Series cells | $N_{ae}$ | 30 |
| Cell area | $A$ | 0.25 m$^2$ |
| Reversible voltage | $U_{rev}$ | 1.229 V |
| *PEMFC Stack* | | |
| Series cells | $N_{pe}$ | 300 |
| Operating temperature | $T_{pe}$ | 323.15 K |
| Membrane area | $A_{pe}$ | 50 cm$^2$ |
| *Battery* | | |
| Nominal voltage | $V_{nom}$ | 240 V |
| Internal resistance | $R_{int}$ | 0.15 Ω |
| Capacity | $C_{Ah}$ | 100 Ah |
| *Battery Bidirectional Converter* | | |
| Inductance | $L_b$ | 2 mH |
| *DC Bus* | | |
| Capacitance | $C_{dc}$ | 30 mF |
| Voltage setpoint | $V_{dc}^*$ | 300 V |
| *DEMPC* | | |
| Sampling interval | $T_s$ | 40 $\mu$s |
| Prediction horizon | $N_p$ | 2–3 |
| Reference horizon | $N_s$ | 10 |